



Advancing dairy farm simulations: A 2-step approach for tailored lactation curve estimation and its systemic impacts

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ABSTRACT

Lactation curve models are a foundational component of dairy farm simulation models because they support prediction of individual animal milk production over time. For farm simulation models to be applicable as decision-support tools, the predicted baseline milk production should match farm reported production as accurately as possible. However, individual animal lactation curve parameters are not easily accessible farm data. This study introduces a straightforward and effective calibration method for determining the parameters of Wood's lactation curve, leveraging a previously published database of parameters and 3 additional data inputs readily available on farms: annual herd milk production (AHMP), number of milking cows, and herd parity composition. Our method involves (1) adjusting curve parameters based on previously reported national estimates and farm contextual metadata (i.e., temporal, geographic, and management factors) and (2) further optimizing the scale parameter for each parity to match the 305-d milk yield derived from observed AHMP, number of milking cows, and herd parity composition. The Ruminant Farm Systems (RuFaS) model, a comprehensive dairy farm simulation platform, was employed to evaluate this method on 10 commercial Holstein dairy farms in New York ($n = 3$), Texas ($n = 3$), and Wisconsin ($n = 4$). By using lactation curve parameters estimated by this calibration method, we achieved greater accuracy in estimating AHMP with RuFaS for these case study farms, reducing the root mean square percentage error from 40.6% to 2.22%. To evaluate the downstream effects of lactation curve estimation methods within a farm systems context, we used the RuFaS Animal Module to simulate key performance and environmental metrics, including dry matter intake, enteric methane production, and manure excretion. We

calculated gross feed efficiency and associated feed and manure emissions using emission factors derived from established literature. The results underscore the critical role of lactation curve modeling in dairy farm system simulation models and its substantial effects on environmental footprint predictions. In conclusion, this study demonstrates the effectiveness of our lactation curve parameter calibration method. With this method, farm system simulation models such as RuFaS can greatly increase their reliability in generating farm-specific predictions without requiring extensive farm data collection. **Key words:** milk production estimation, farm-specific Wood's curve parameters, dairy farm simulation, RuFaS, optimization

INTRODUCTION

The Ruminant Farm Systems (RuFaS; <https://github.com/RuminantFarmSystems/RuFaS>) model is a cutting-edge dairy farm system model developed by a nationwide team of scientists and engineers to support on-farm decision-making and farm systems research by simulating a comprehensive range of processes, including crop growth, feed storage, animal life cycles, and manure management (Kebreab et al., 2019). It operates based on user input data, incorporating variables such as weather conditions, farm infrastructure, animal attributes, and management practices. The model's Animal Module outputs daily, granular data related to herd dynamics, including the average number of animals at each life stage and the herd's parity composition. Moreover, it provides estimates for key production indicators such as feed intake and milk yield, as well as environmental metrics such as manure excretion and GHG emissions. Recent publications have highlighted the RuFaS Animal Module's capabilities, demonstrating its potential to inform decision-making across various aspects of dairy farm management (Hansen et al., 2021; Reed, 2021; Li et al., 2023b).

An essential component of RuFaS and other dairy farm simulation models is a lactation curve model, a

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nonlinear mathematical equation that describes the milk production pattern of dairy cows over time, specifically time since parturition (Macciotta et al., 2011). Complex lactation curve models offer greater flexibility by adapting to deviations in milk production caused by factors such as disease and management (Val-Arreola et al., 2004; Cole et al., 2012). However, these models often require extensive data to calibrate, making it challenging to parameterize them in data-limited situations. In contrast, the Wood's lactation curve model (Wood, 1967), a simpler, still robust and widely used approach, is more adaptable for such contexts and was incorporated into RuFaS to determine parity-specific average lactation curves. The Wood's lactation curve model is expressed as

$$y = at^be^{-ct} + \varepsilon, \quad [1]$$

where y represents the observed (or predicted) daily milk yield (kg) at day t , t denotes the number of DIM, a is a scale factor for production level, b indicates the growth rate in milk yield until peak production, controlling the shape of the ascending phase of the lactation curve, c describes the rate of decline after the peak, controlling the shape of the descending phase of the lactation curve, and ε represents the normally distributed random residual (Wood, 1967; Li et al., 2022).

In RuFaS, we assume that each lactation curve parameter (a , b , c) for a specific herd, and for each parity, follows its own normal distribution with a distinct mean and variance. The RuFaS model's Animal Module is stochastic (Hansen et al., 2021; Li et al., 2023b), and it randomly assigns lactation curve parameters to each lactating animal at the beginning of each lactation, based on the distribution of these parameters. As a result of this randomization, animals within the same parity group in a single simulation can have different lactation parameters. Each animal retains its assigned parameters until the end of its current lactation and is assigned a new, independent set of parameters upon entering the next lactation.

Currently, RuFaS uses the method described by Li et al. (2022) to determine the parity-specific mean and variance of Wood's lactation curve parameters. Based on a comprehensive dataset from the US Council on Dairy Cattle Breeding, which included over 8 million Holstein cow lactations across 21,056 herds, these authors derived national average parameter values for the Wood's lactation model: $a = 19.9$, $b = 0.247$, and $c = 0.003376$ (Li et al., 2022). They further developed adjustments for contextual factors, including spatial elements such as the farm's location, temporal aspects such as calving year and month, and management variables such as milking frequency and parity (Li et al., 2022). These adjustments, calculated using a mixed-effect model (Li et al., 2022),

refined the averages to more accurately represent lactation dynamics under specific farm conditions.

Despite accounting for spatial, temporal, and management factors, the method established by Li et al. (2022) has critical limitations. It overlooks other farm-level influences on milk production, making it fall short for farm-specific use cases. For example, even when 2 farms of the same size share similar locations, milking frequencies, parity compositions, and calving year and month, they may still exhibit substantial differences in annual herd milk production (AHMP). These differences can arise from variations in genetic profiles, diets, reproductive programs, culling decisions, and disease treatment protocols, and underscore the need for a more tailored approach to accurately determine farm-specific lactation curve parameters.

Accurate estimation of these parameters is essential to ensuring the precision and effectiveness of RuFaS predictions for animal- and farm-scale productivity and environmental outcomes. The milk production dynamics predicted by the Wood's lactation curve model influence numerous intermediate outputs and overall farm performance predicted by RuFaS. For example, the day of dry-off and the total length of each lactation cycle are determined by a combination of the lactation curve parameters and the user-defined low milk production threshold, below which cows are dried off and enter the dry period. These factors, in turn, influence RuFaS predictions for animal and herd productivity, feed intake, enteric methane (CH₄) production, and manure excretion and emissions. Although existing literature often examines the effects of lactation curves and their parameters in isolation (Buttchereit et al., 2010; Chen et al., 2022; Chiapella and Garcia, 2022), it is important to evaluate their role within the broader context of a whole dairy farm system, particularly their downstream impacts on environmental footprints.

The livestock sector contributes 12% of human-induced global GHG emissions, with cattle (beef and dairy) responsible for 62% of the sector emissions, equating to 3.8 gigatons of CO₂ equivalents (CO₂-eq) annually (Capper and Cady, 2020; FAO, 2023). Within dairy farming, the primary contributors to GHG emissions are enteric fermentation (CH₄), manure management and application (N₂O, CH₄), and feed production (CO₂, N₂O), constituting 47%, 26%, and 18% of total emissions during milk production, respectively (Gerber et al., 2013). Modeling tools that estimate farm-scale emissions can support regional and national inventories to help track emissions over time and identify management strategies that can reduce whole-farm emissions while maintaining productivity (Veltman et al., 2017; Hansen et al., 2021).

To increase the accuracy and applicability of whole-farm models such as RuFaS, we propose a calibration

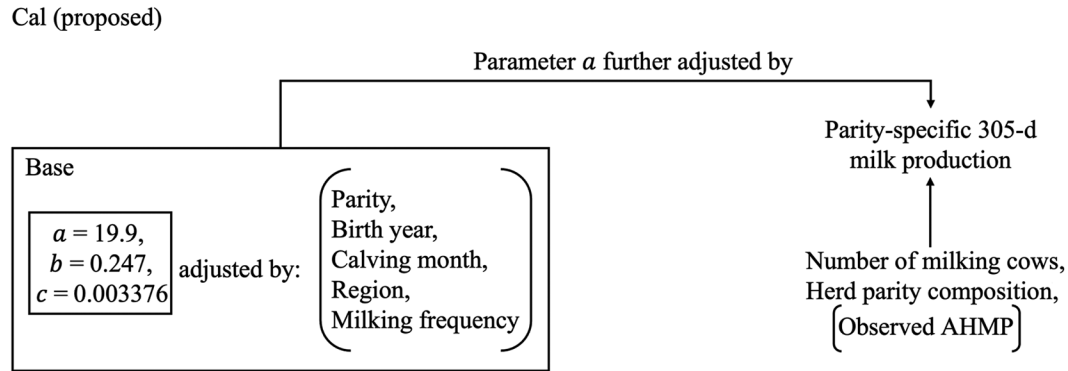


Figure 1. Graphical illustration of the baseline (Base) and calibrated (Cal) methods for determining the Wood's lactation curve parameters. Parameter a is the scale factor for the production level, b indicates the growth rate in milk yield until peak production, and c describes the rate of decline after the peak (Wood, 1967). Variables in the brackets are inputs from users. The Base method uses average parameters estimated in a previous study (Li et al., 2022). In our case study, the number of milking cows and herd parity composition were estimated using the Ruminant Farm Systems (RuFaS) model, with lactation curve parameters estimated from the Base method due to lack of available data. However, users can input actual values for these variables. AHMP = annual herd milk production.

method based on the approach developed by Li et al. (2022) for defining Wood's lactation curve parameters. Specifically, this study has 2 main objectives: (1) to present and evaluate a method for defining and calibrating farm-specific Wood's lactation curve parameters using minimal farm data; and (2) to assess the influence of this method on downstream model outputs, such as feed intake, manure production and composition, and GHG emissions, using the RuFaS model's Animal Module.

MATERIALS AND METHODS

This study introduces a method to calibrate whole-farm milk production estimates from the RuFaS model to farm-specific production values using parameters from the Wood's lactation model. Our method relies on customization of parameters by parity number, region, calving year, month, and milking frequency previously reported by Li et al. (2022) and further calibrating the scale parameter based on parity-specific 305-d milk production (M305) derived from observed farm's AHMP, number of milking cows, and herd parity composition. We selected AHMP as the metric of farm-specific milk production due to its high accessibility, intuitive relevance, and clear correlation with dairy farmers' financial returns. Unlike other farm-level metrics, such as test-day milk yield, which represents a single point in time, AHMP includes a full year of data, making it both practical and relevant for farm-specific applications.

Parameter Definition

The current method in RuFaS for obtaining lactation curve parameters is referred to as the baseline (Base) approach, where parameter values are adjusted for

temporal, geographic, and management factors (Li et al., 2022). Our proposed calibration method, labeled as calibrated (Cal), builds upon the Base by incorporating an additional farm-specific adjustment for parameter a , using observed AHMP, number of lactating cows, and herd parity composition. Figure 1 graphically illustrates the relationship between the 2 methods, with the Base relying on national averages and farm metadata, and the Cal further refining scale parameter a with farm-specific data.

Ideally, observed values for AHMP, number of lactating cows, and herd parity composition should all be available and align with the input farm management strategies. However, in cases where the number of lactating cows and parity composition are unavailable—as was true in our study—the expectation of these values can be predicted by the RuFaS model based on the provided farm management inputs and the lactation curve parameters estimated by the Base. This approach assumes that the current herd is at steady state under the given management conditions and that the parity composition does not differ significantly between the 2 lactation curve estimation methods, which is expected because the herd parity composition is primarily influenced by reproduction performance and culling strategy. To further validate this assumption, we compared the herd parity composition simulation results using both methods.

To reduce farm data requirements for simulation model application, the Cal method deliberately focuses on adjusting the scale parameter a , rather than attempting to fine-tune all 3 parameters (a , b , and c). We employ a straightforward mathematical procedure to estimate 3 distinct values of parameter a , one each for cows in parity 1, parity 2, and parity 3 and beyond. These estimates of a are the values that produce the predicted M305 that most

closely matches the observed data for each parity group. The first step is to calculate the average M305 for each parity using the observed AHMP, along with the input or estimated number of lactating cows and herd parity composition according to Equations 2 through 5. Specifically, Equation 2 calculates the herd's average M305 using the observed AHMP, and Equations 3 through 5 solve the parity-specific M305 ($M305_{P1}$, $M305_{P2}$, and $M305_{P3+}$) using the herd's average M305 and the average population percentage of each parity group:

$$M305_{herd} = \frac{AHMP_{observed}}{N_{milking} \times 365} \times 305. \quad [2]$$

Here, $M305_{herd}$ is the herd's average 305-d milk production (kg/cow per 305 d), $AHMP_{observed}$ is the observed annual herd milk production (kg/yr), and $N_{milking}$ is the average number of lactating cows in the herd every day.

$$M305_{herd} = M305_{P1} \times P1\% + M305_{P2} \times P2\% + M305_{P3+} \times P3^+, \quad [3]$$

$$M305_{P2} = M305_{P1} \times 118\%, \quad [4]$$

$$M305_{P3+} = M305_{P1} \times 125\%. \quad [5]$$

Here, $M305_{P1}$, $M305_{P2}$, and $M305_{P3+}$ are the average 305-day milk production (kg/cow per 305 d) for cows in parity 1, parity 2, and parity 3 and beyond, respectively; P1%, P2%, and P3+% are the population percentages of the corresponding parity groups, respectively, either input by users or estimated based on the RuFaS simulation. Equations 4 and 5 represent the relationships between average M305 of parity 1 and that of parity 2 and parity 3 and beyond, based on the work of Li et al. (2022). Using the relationships established in Equations 3 through 5, we first solve for $M305_{P1}$ by substituting $M305_{P2}$ and $M305_{P3+}$ in Equation 3 for the alternative definitions for these variables presented in Equations 4 and 5 based on $M305_{P1}$. With the result for $M305_{P1}$, we apply Equations 4 and 5 to calculate $M305_{P2}$ and $M305_{P3+}$.

After obtaining the observed parity-specific M305 values, we can fit parameter a to each parity's average M305. This fine-tuning is achieved through an iterative adjustment process, known as grid search (Bergstra and Bengio, 2012). We vary parameter a by small increments (0.01) within a range from -20 to $+20$, relative to the value obtained from the initial adjustment using the Base. The bound of ± 20 was chosen to ensure $\pm 100\%$ coverage of the national average of 19.9. For each possible value of parameter a within this range, the M305 is recalculated using Equation 6:

$$M305 = \int_1^{305} at^b e^{-ct} dt, \quad [6]$$

where $M305$ is the 305-d milk production (kg/cow per 305 d); a , b , and c are lactation curve parameters; and t is the DIM. This recalculated $M305$ is then compared to the observed parity-specific $M305$ obtained previously, and the absolute difference between the two is recorded. By iterating through all possible adjustments within the pre-defined range, we identify the adjustment to parameter a that minimizes this absolute difference. The process can be summarized as Equation 7:

$$a^* = \operatorname{argmin}_{a \in [a_{min}, a_{max}]} |M305_{calculated} - M305_{observed}|, \quad [7]$$

where the argmin refers to the value of parameter a within $[a_{min}, a_{max}]$ that minimizes the absolute difference between the calculated and observed $M305$, denoted as a^* , the optimal lactation curve parameter. a_{min} and a_{max} are the parameters obtained when we subtract or add 20 from the estimate of a from the Base, respectively.

Our method is implemented in Python (version 3.10, Python Software Foundation, <https://www.python.org/>) as part of the RuFaS model. Additionally, it is available as a standalone version on GitHub (<https://github.com/YijingGong/Herd-specific-lactation-curve.git>). The GitHub repository allows users to calculate farm-specific lactation curve parameters using either the Base or the Cal method. Detailed instructions for generating these parameters are provided in the repository's README file.

Impact Assessment with Case Study Farms

To assess the impact of our method for calibrating farm-specific lactation curve parameters, we simulated 10 Holstein dairy farms in the US with the RuFaS model using lactation curve parameters defined by the Base and the Cal methods. Then, we compared the observed with the AHMP estimates. Other production outcomes, including feed intake and manure excretion, and environmental outcomes such as GHG emissions were also calculated to understand the downstream impacts. Due to the stochastic nature in the RuFaS model's Animal Module, we ran the model 4 times (replications) for each set of inputs, and then reported the average results of these simulation runs.

Input Definition. To obtain the required RuFaS model inputs and observed parity-specific M305, we collected data from 10 commercial Holstein dairy farms (F1–F10) across 3 US states: New York (NY, $n = 3$: F1, F2, F3), Texas (TX, $n = 3$: F4, F5, F6), and Wisconsin (WI, $n = 4$: F7, F8, F9, F10) in 2023. Table 1 provides

descriptive statistics for each farm, including region, number of adult cows, AHMP, average daily milk production, milking frequency, reproductive program, cow conception rate, culling milk production threshold, “do-not-breed (DNB)” time, and average mature body weight. The descriptive statistics in Table 1 provide the key inputs for the RuFaS Animal Module (Hansen et al., 2021; Li et al., 2023b).

To derive lactation curve parameters, we used the same calving year (2016) for both methods, because 2016 represents the most current data available from research by Li et al. (2022). Further, because our objective is to estimate annual milk production, we did not designate a specific calving month. This decision is supported by the evidence that the effects of seasonal variations on lactation curve parameters tend to neutralize over the course of a year (Li et al., 2023a). Parity composition was derived by running RuFaS simulations using farm-provided management inputs and the Base method for 2 reasons. First, parity composition data were not collected from the case study farms. Second, even if the data are collected, a single snapshot of herd composition may not reflect a stabilized parity structure that accounts for a farm’s management strategies (e.g., reproduction or culling). In contrast, using the RuFaS model allows us to simulate a herd parity structure that is consistent with ongoing farm management practices. Further, because the only variable changed in our evaluation is the lactation curve parameter estimation method (Base versus Cal), use of the simulated parity compositions in the calibration method isolates the effects of the study to the lactation curve parameterization. To confirm that the parity distributions were the same between methods, we compared the herd parity compositions predicted by RuFaS using lactation curve parameters from both the Base and the Cal methods.

In addition to the farm-specific herd management inputs described in Table 1, we developed national or regional default values for the remaining inputs required by the RuFaS model. The regional diets used in this study were previously formulated by industry experts to capture average diets in each region using the Nutritional Dynamic System ration formulation software (RUM&N Sas, Reggio Emilia, Italy) based on the Cornell Net Carbohydrate and Protein System (CNCPS version 6.55; Van Amburgh et al., 2015). Diets were formulated to deliver the nutrients necessary to meet the requirements for 110% of the regional average milk production per cow per day reported by USDA for the year 2020 (Appendix Table A1). The feeds included in the diets for each region were based on data collected from over 2,000 farms through a survey conducted by the FARM-ES group and complemented by findings from recent studies (Asselin-Balencçon et al., 2013; Thoma et al., 2013; de Ondarza

Table 1. Region, number of adult cows, observed annual herd milk production (AHMP), average daily milk production, milking frequency, reproductive program, cow conception rate, culling milk production threshold, “do-not-breed (DNB)” time, and average mature body weight of the 10 US Holstein dairy farms used to assess the impacts of lactation curve parameter estimation methods

Farm ID	Region	Adult cow number	Observed AHMP (kg/yr)	Average daily milk production (kg/cow per day)	Milking frequency (times/d)	Cow reproductive program ¹	Cow conception rate (%)	Culling milk production threshold ² (kg)	DNB time ³ (d)	Average mature body weight (kg/cow)
F1	NY	1,500	1.84×10^7	33.6	3	TAI	48	22.7	275	794
F2	NY	50	5.76×10^5	31.6	2	ED	38	20.0	300	612
F3	NY	1,002	1.12×10^7	30.6	3	ED-TAI	28	29.5	300	635
F4	TX	3,650	3.92×10^7	29.5	2	ED-TAI	49	27.2	225	612
F5	TX	2,450	3.38×10^7	37.5	3	ED-TAI	36	24.9	200	590
F6	TX	8,202	9.93×10^7	33.2	2	ED-TAI	38	31.7	250	740 ⁴
F7	WI	58	4.26×10^5	20.1	2	ED-TAI	55	13.6	225	544
F8	WI	2,796	4.09×10^7	40.1	3	ED-TAI	53	31.8	225	816
F9	WI	80	4.88×10^5	16.7	2	ED-TAI	35	22.7	200	748
F10	WI	385	5.31×10^6	37.8	3	TAI	45	22.7	300	740 ⁴

¹TAI = timed artificial insemination; ED = estrus detection.

²Cows marked as “DNB” are culled if their milk production drops below this threshold. Culling details were described by Li et al. (2023b).

³If cows remain nonpregnant for an extended period after calving, they are marked as “DNB.” Culling details were described by Li et al. (2023b).

⁴Data not reported, so the default value of 740 kg (Li et al., 2023b) was used for simulations.

and Tricarico, 2021). The dietary nutrient compositions of the regional diets are presented in Appendix Table A2. A database of emissions factors was compiled from 3 sources. County-specific emissions factors for 7 of the most commonly used dairy feeds (alfalfa hay, alfalfa haylage, corn grain, corn silage, dried distillers grains, soybean meal, wheat middlings) were sourced from the Food System Supply-chain Sustainability (**FoodS³**) model (Pelton et al., 2021, 2024; <https://foodscubed.umn.edu/>). Lifecycle Environmental Impact Footprinting consulting (Minneapolis, MN), in coordination with collaborators from the University of Minnesota FoodS³ group, was commissioned to estimate regionally specific emission factors for 17 commonly fed by-products (almond hulls, brewers grains, canola meal, cereal waste, citrus pulp, corn cannery waste, wet corn distillers grains, dry corn gluten feed, wet corn gluten feed, whole cottonseed, malt sprouts, cane molasses, soybean hulls, defatted soybean meal, acid whey, condensed whey, and powdered whey). National averages emissions factors for the remaining feeds were sourced from the Intergovernmental Panel on Climate Change (IPCC, 2023).

Outcome Assessment. To assess the accuracy of the estimated AHMP, we calculated the percentage of observed AHMP:

$$\% \text{ of } AHMP_{\text{observed}} = \left(\frac{AHMP_{\text{simulated}}}{AHMP_{\text{observed}}} \right) \times 100. \quad [8]$$

Here, $AHMP_{\text{simulated}}$ is the simulated annual herd milk production (kg/yr).

In addition to milk production, we reported other production outcomes from the Animal Module, including DMI, manure nitrogen, and manure degradable and non-degradable volatile solids. Gross feed efficiency (**GFE**) was then calculated using DMI and fat- and protein-corrected milk (**FPCM**; 3.5% fat, 3.2% crude protein) as follows:

$$GFE = \frac{FPCM}{DMI}, \quad [9]$$

where $FPCM$ (kg/yr) is fat- and protein-corrected milk, calculated as $FPCM \text{ (kg/yr)} = 0.327 \times \text{milk production (kg/yr)} + 7.2 \times \text{milk protein yield (kg/yr)} + 12.95 \times \text{milk fat yield (kg/yr)}$, and DMI is dry matter intake (kg/yr; DRMS, 2014; Hall, 2023).

Reported environmental outcomes include enteric CH_4 emissions, feed CO_2 -eq emissions, and manure CH_4 and N_2O emissions. Enteric CH_4 emissions was calculated using the equations by Niu et al. (2018) in the Animal Module. Total feed CO_2 -eq emissions were manually calculated using the feed emission factors (kg

CO_2 -eq/kg DMI) and the predicted DMI (kg/yr). Because GHG emissions from manure management vary widely depending on manure management systems and weather conditions (IPCC, 2019), we used regional emissions factors for a single representative manure management system (liquid or slurry pit storage) to illustrate the potential downstream effects of lactation curve inputs on manure emissions without the confounding effects of different manure management systems. According to the IPCC (2019), the manure CH_4 emission factors for a liquid or slurry pit storage system were 33.8, 33.8, and 119.0 g CH_4 /kg manure volatile solids for farms in New York, Wisconsin, and Texas, respectively, due to differing weather conditions. For manure N_2O emissions, the emission factor used for the pit storage system was 0.002 kg N_2O /kg N excreted for all farms. All GHG emissions were converted to CO_2 -eq to compute the whole-farm GHG emissions, using the 100-year Global Warming Potential values of 27 for CH_4 and 273 for N_2O (IPCC, 2023).

Statistical Analysis. We assessed the accuracy of simulated AHMP against observed AHMP using 2 statistical metrics. Specifically, we computed the mean absolute percentage error (**MAPE**; Equation 10) and the root mean square percentage error (**RMSPE**; Equation 11). Both metrics express errors in percentage terms, which allows for a fair comparison across farms with widely varying production levels:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i^{\text{pred}} - y_i^{\text{obs}}}{y_i^{\text{obs}}} \right| \times 100\%; \quad [10]$$

$$RMSPE = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{y_i^{\text{pred}} - y_i^{\text{obs}}}{y_i^{\text{obs}}} \right)^2} \times 100\%. \quad [11]$$

Here, y_i^{obs} and y_i^{pred} denote the observed and predicted values of the variable ($AHMP$) at instance i , respectively. The index i ranges from 1 to n , where n is the total number of observations. To determine significant differences in the accuracy of AHMP estimates between the Base and the Cal methods, we performed a paired Wilcoxon signed-rank test on the absolute differences between the simulated and observed AHMP. Additionally, to assess the assumption about the herd parity composition, we applied a mixed-effects model to analyze the herd parity composition generated by the 2 methods, treating the method (Base or Cal) as a fixed effect and the farm as a random effect. In this study, a P -value less than 0.05 was considered significant. All statistical analyses were performed using the `scipy.stats` package in Python (Virtanen et al., 2020).

Table 2. Summary of statistical metrics assessing the performance of the baseline (Base) and calibrated (Cal) methods in accurately estimating annual herd milk production

Item ¹	Base	Cal
MAPE	23.8%	2.04%
RMSPE	40.6%	2.22%

¹MAPE = mean absolute percentage error; RMSPE = root mean square percentage error.

RESULTS AND DISCUSSION

Herd Demographics

Herd sizes of the case study farms ranged from 50 to 8,202, and AHMP ranged from 426,295 kg/yr to 99,336,729 kg/yr. The average milk production per cow across these farms ranged from 6,104 to 14,611 kg/cow per year. Their simulated parity compositions are shown in Appendix Table A3. The average simulated distributions of parity 1, 2, and 3+ cows across the 3 NY farms were 31.0%, 22.9%, and 46.1%, respectively. For the TX farms, the corresponding figures were 42.2%, 27.1%, and 30.7%, and the WI farms showed average distributions of 36.5%, 24.8%, and 38.7%. Analysis using the mixed-effects model revealed no statistically significant differences in the simulated parity structure between the outcomes generated by the Base and the Cal ($P > 0.05$). In addition, these simulated parity compositions closely align with national average values for parity distributions provided by a Dairy Records Management Systems (Raleigh, NC) benchmarking tool, DairyMetrics. Based on data from 6,041 herds in the United States, DairyMetrics reported average percentages of 34.6%, 27.2%, and 37.9% for cows in parities 1, 2, and 3+, respectively, as of August 2024 (DRMS, 2024).

Although observed parity data were unavailable for the case study farms, our analysis supports using the parity composition estimated by the Base method as a valid input for the Cal method. This conclusion is grounded in the mechanistic structure of the RuFaS Animal Module, which links parity composition to key drivers such as reproductive program, conception rate, parity-specific culling thresholds, and nonproduction culling rates (Li et al., 2023b). The only functional difference between the Base and Cal methods lies in the lactation curve parameters. Although these parameters affect the DIM at which cows reach the culling threshold, they do not substantially influence the parity structure, as they alter the timing of culling rather than the decision to cull itself. Assuming the RuFaS model accurately reflects biological processes (Hansen et al., 2021; Li et al., 2023b), the nonsignificant differences in simulated parity distributions suggest that the model has stabilized toward an expected structure under the given input management strategies, regardless

of the lactation curve estimation method (Base or Cal). Therefore, we consider it appropriate to use the simulated parity composition from the Base method as an input to the Cal method for the purpose of evaluation of the 2 methods for defining lactation curve inputs for application of the RuFaS model. Additionally, we emphasize the flexibility of our approach: when actual parity composition data are available, users can input these values directly. In such cases, parity-specific M305 values will be calculated according to the user inputs (Equations 3–5), and the Cal method will then adjust the lactation curve parameters based on these calculated parity-specific M305 (Equation 7).

Effect of Proposed Method for Lactation Curve Parameter Definition

Figure 2 graphically illustrates the percentage of observed AHMP (ratio of $AHMP_{simulated}$ to $AHMP_{observed}$), providing an intuitive way to compare the performance of the 2 methods. Base generally produced AHMP estimates that fell within 80% to 120% of the observed values, with 2 notable outliers: F7 (167%) and F9 (206%). In contrast, Cal showed visible improvements, as expected, with all farms achieving AHMP estimates within $\pm 4\%$ of the observed values. Across the 10 farms, Cal improved the percentage of observed AHMP by an average of 22% and a median of 7%, with a range from 0% to 104%. The evaluation metrics presented in Table 2 further confirmed a higher accuracy of Cal. The Cal method achieved a substantially lower MAPE (2.04% vs. 23.8%) and RMSPE (2.22% vs. 40.6%). The paired Wilcoxon signed-rank test confirmed that the absolute differences between simulated and observed AHMP were significantly different ($P = 0.009$) between Base and Cal, underscoring the importance of calibration for obtaining accurate farm-specific lactation curves.

The effectiveness of the Base method largely depended on how closely a farm's milk production aligned with the average performance of farms in the data set used by Li et al. (2022). For instance, farms such as F1 and F4, with average individual cow milk production of 12,270 kg/cow per year and 12,839 kg/cow per year, respectively, closely aligned with their regional averages (e.g., the New York state average of 12,721 kg/cow per year; CDCB, 2024). In contrast, farms such as F9 (6,103 kg/cow per year) and F7 (7,351 kg/cow per year) exhibited substantial deviations from the Wisconsin state average of 13,340 kg/cow per year (CDCB, 2024). These deviations contributed to the suboptimal performance of Base for such farms. In contrast, the Cal method, by incorporating one additional data point (parity-specific M305) to adjust parameter a , ensured that simulated AHMP estimates remained within a narrow deviation from observed

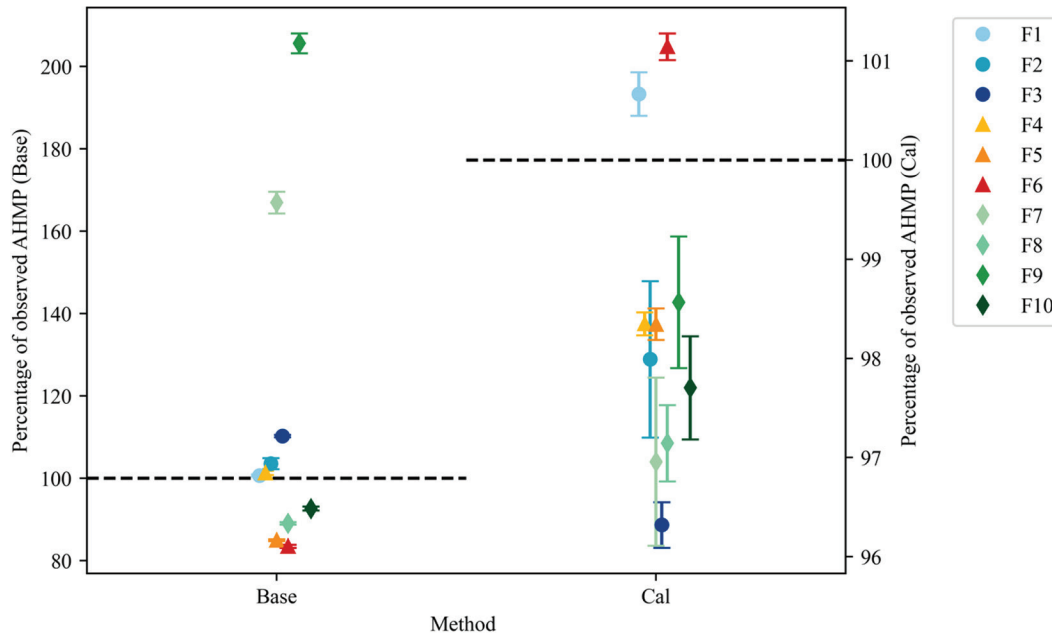


Figure 2. The Ruminant Farm Systems (RuFaS) model simulated annual herd milk production (AHMP), expressed as percentage of observed AHMP using the baseline (Base) and calibrated (Cal) methods for determining lactation curve parameters for 10 US dairy farms in the states of New York (circles), Texas (triangles), and Wisconsin (diamonds). The Base method adjusts Wood's lactation curve parameters based on farm-specific metadata (i.e., temporal, geographic, and management factors; Li et al., 2022). The Cal method, our proposed method, further refines the scale parameter a by incorporating observed data on annual herd milk production, number of milking cows, and herd parity composition. The error bars represent SD.

AHMP, making Cal a robust method for simulating individual farm performance despite the large variability in milk production across farms. Currently, the RuFaS model defaults to the Base method, which limits its reliability for simulating specific farms. This underscores the importance of adopting the Cal method to improve the accuracy and robustness of RuFaS simulations.

Another intrinsic limitation of Base is its reliance on historically fixed datasets (data up to 2016) and its lack of ability to continuously update the lactation curve parameter dataset. The current dataset does not account for the ongoing genetic and management improvements in dairy cattle that increase milk yield over time. Updating the analysis performed by Li et al. (2022) is time consuming and limited by resources and data accessibility. A time series decomposition of over 8 million lactation curves (Li et al., 2023a) indicated that annual milk yield has risen by 1.15% per year on average. Using outdated datasets may not accurately reflect current cow performance. Even after extrapolating the Base simulated AHMP to account for an 8.05% increase over 7 years since 2016, the method still produced a widely scattered percentage of observed AHMP, ranging from 90% to 120%, excluding 2 outliers. A second limitation of Base is that it can only predict the average AHMP for a given region and set of management practices and ignores variability across dairy farms that occurs due to differences in management

strategies, diets, starting genetics, and rate of genetic improvement. Therefore, a uniform linear extrapolation for all farms does not suffice. In contrast, Cal adjusts parameter a to account for some of these intrinsic variabilities across farms, resulting in a closer match to the observed AHMP.

Effects on Production and Environmental Outcomes

The production performance indicators, including milk production, DMI, GFE, and manure excretion of lactating cows, are detailed in Table 3. Environmental footprint indicators, including GHG emissions from enteric fermentation, feed, and manure, are shown in Table 4. Given the close relationship between production and environmental indicators and milk production level, and the improved accuracy of Cal in estimating AHMP, the estimates of other performance and environmental indicators from Cal should be more representative of the simulated herd.

Dry matter intake estimates from Cal ranged from 19.5 to 29.1 kg/cow per day, closely aligning with the average intake range of 20.8 to 29.4 kg/cow per day reported by NASEM (2021). Gross feed efficiency estimates derived from Cal exhibited uniformity across New York farms (F1–F3) and were all close to 1.35. In contrast, Texas (F4–F6) and Wisconsin (F7–F10) farms showed more variation, with GFE ranging from 1.53 to 1.80 and

Table 3. Production performance indicators predicted by the Ruminant Farm Systems (RuFaS) model using lactation curve parameters determined by the baseline (Base) and calibrated (Cal) methods for 10 US Holstein dairy farms in the states of New York (F1–F3), Texas (F4–F6), and Wisconsin (F7–F10)

Farm	Method ¹	Daily milk production (kg/cow per day)		Dry matter intake (kg/ cow per day)		Gross feed efficiency (kg/ kg)		Manure nitrogen (kg/cow per day)		Manure degradable volatile solids (kg/cow per day)		Manure nondegradable volatile solids (kg/cow per day)	
		Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
F1	Base	39.6	0.09	29.1	0.09	1.38	0.002	0.508	0.0018	9.31	0.029	1.05	0.003
	Cal	39.5	0.06	29.1	0.05	1.38	0.000	0.506	0.0008	9.30	0.017	1.05	0.002
F2	Base	36.6	0.58	25.8	0.27	1.35	0.008	0.439	0.0036	8.21	0.094	0.934	0.0100
	Cal	34.7	0.37	24.7	0.13	1.34	0.009	0.422	0.0018	7.80	0.045	0.890	0.0048
F3	Base	37.6	0.08	23.5	0.26	1.67	0.019	0.409	0.0044	7.38	0.087	0.844	0.0091
	Cal	32.7	0.03	25.1	0.02	1.36	0.001	0.424	0.0006	7.98	0.006	0.911	0.0006
F4	Base	34.4	0.05	23.3	0.05	1.37	0.001	0.425	0.0008	7.32	0.016	0.840	0.0016
	Cal	33.3	0.02	21.4	0.05	1.53	0.004	0.390	0.0009	6.65	0.017	0.768	0.0018
F5	Base	36.2	0.11	22.2	0.02	1.67	0.005	0.405	0.0004	6.93	0.007	0.797	0.0007
	Cal	41.5	0.05	23.6	0.04	1.80	0.002	0.429	0.0007	7.39	0.013	0.845	0.0013
F6	Base	35.1	0.02	26.1	0.02	1.38	0.001	0.476	0.0003	8.24	0.008	0.937	0.0008
	Cal	42.1	0.02	26.2	0.01	1.65	0.001	0.477	0.0002	8.25	0.003	0.936	0.0003
F7	Base	37.8	0.55	26.8	0.24	1.36	0.009	0.445	0.0051	8.55	0.079	0.968	0.0083
	Cal	22.0	0.27	20.0	0.31	1.06	0.028	0.339	0.0049	6.23	0.103	0.722	0.0107
F8	Base	40.7	0.03	28.8	0.42	1.48	0.022	0.482	0.0071	9.20	0.146	1.03	0.015
	Cal	44.3	0.11	26.6	0.02	1.75	0.003	0.444	0.0003	8.42	0.007	0.950	0.0010
F9	Base	37.9	0.10	27.8	0.70	1.44	0.040	0.469	0.0118	8.89	0.243	1.00	0.026
	Cal	18.4	0.27	19.5	0.12	0.997	0.0120	0.332	0.0020	6.07	0.041	0.705	0.0043
F10	Base	40.7	0.08	28.6	0.39	1.40	0.021	0.483	0.0068	9.15	0.135	1.03	0.014
	Cal	42.8	0.19	25.0	0.46	1.68	0.030	0.420	0.0075	7.92	0.157	0.898	0.0165

¹The Base method adjusts Wood's lactation curve parameters based on farm-specific metadata (i.e., temporal, geographic, and management factors; Li et al., 2022). The Cal method further refines the parameter *a* (scale factor for production level) by incorporating observed data on annual herd milk production, number of milking cows, and herd parity composition.

Table 4. Environmental footprint indicators estimated by the Ruminant Farm Systems (RuFaS) model using lactation curve parameters determined by the baseline (Base) and calibrated (Cal) methods for 10 US Holstein dairy farms in the states of New York (F1–F3), Texas (F4–F6), and Wisconsin (F7–F10)

Farm	Method ¹	Enteric CH ₄ (kg/cow per day)		Feed CO ₂ (kg/cow per day)		Manure CH ₄ (kg/cow per day)		Manure N ₂ O (kg/cow per day)		Total GHG production (kg CO ₂ -eq/cow per day)		Total GHG intensity (kg CO ₂ -eq/kg milk)	
		Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
F1	Base	0.493	0.0013	15.1	0.05	0.350	0.0011	0.00160	5.79 × 10 ⁻⁶	38.3	0.11	0.969	0.0014
	Cal	0.493	0.0007	15.1	0.03	0.350	0.0006	0.00159	2.60 × 10 ⁻⁶	38.3	0.06	0.969	0.0002
F2	Base	0.448	0.0029	13.4	0.14	0.309	0.0035	0.00138	1.15 × 10 ⁻⁵	34.2	0.31	0.935	0.0069
	Cal	0.434	0.0024	12.8	0.07	0.294	0.0017	0.00133	5.56 × 10 ⁻⁶	32.8	0.16	0.947	0.0072
F3	Base	0.428	0.0035	12.2	0.13	0.278	0.0033	0.00129	1.37 × 10 ⁻⁵	31.6	0.32	0.841	0.0088
	Cal	0.449	0.0004	13.0	0.01	0.300	0.0002	0.00133	1.82 × 10 ⁻⁶	33.6	0.02	1.03	0.001
F4	Base	0.524	0.0011	12.1	0.03	0.972	0.0021	0.00134	2.50 × 10 ⁻⁶	52.9	0.11	1.54	0.001
	Cal	0.389	0.0006	11.1	0.03	0.883	0.0023	0.00123	2.76 × 10 ⁻⁶	45.8	0.11	1.37	0.004
F5	Base	0.404	0.0005	11.6	0.01	0.919	0.0009	0.00127	1.24 × 10 ⁻⁶	47.6	0.05	1.32	0.004
	Cal	0.420	0.0006	12.3	0.02	0.980	0.0017	0.00135	2.17 × 10 ⁻⁶	50.4	0.08	1.21	0.001
F6	Base	0.458	0.0002	13.6	0.01	1.09	0.001	0.00150	9.69 × 10 ⁻⁷	55.8	0.05	1.59	0.001
	Cal	0.458	0.0001	13.6	0.01	1.09	0.0004	0.00150	5.53 × 10 ⁻⁷	55.9	0.02	1.33	0.001
F7	Base	0.463	0.0024	12.7	0.11	0.322	0.003	0.00140	1.61 × 10 ⁻⁵	34.3	0.26	0.907	0.0074
	Cal	0.383	0.0059	9.50	0.15	0.235	0.0038	0.00107	1.54 × 10 ⁻⁵	26.5	0.42	1.20	0.031
F8	Base	0.499	0.005	13.7	0.2	0.346	0.0055	0.00152	2.24 × 10 ⁻⁵	36.9	0.49	0.906	0.0120
	Cal	0.473	0.0002	12.6	0.01	0.317	0.0003	0.00140	1.07 × 10 ⁻⁶	34.3	0.02	0.775	0.0014
F9	Base	0.486	0.0088	13.2	0.33	0.334	0.0091	0.00147	3.70 × 10 ⁻⁵	35.8	0.83	0.943	0.0239
	Cal	0.391	0.002	9.27	0.06	0.229	0.0015	0.00104	6.20 × 10 ⁻⁶	26.3	0.16	1.43	0.018
F10	Base	0.479	0.0047	13.6	0.19	0.344	0.005	0.00152	2.14 × 10 ⁻⁵	36.2	0.46	0.890	0.0123
	Cal	0.437	0.0055	11.9	0.22	0.298	0.0059	0.00132	2.37 × 10 ⁻⁵	32.1	0.53	0.751	0.0125

¹The Base method adjusts Wood's lactation curve parameters based on farm-specific metadata (i.e., temporal, geographic, and management factors; Li et al., 2022). The Cal method further refines the parameter *a* (scale factor for production level) by incorporating observed data on annual herd milk production, number of milking cows, and herd parity composition.

0.997 to 1.75, respectively. Hutjens (2005) recommended the GFE—defined as the ratio of fat-corrected milk to DMI—for lactating cows between 1.4 and 1.8, with observed values ranging from 1.1 to 2.0. Similarly, Seymour et al. (2020) reported a mean GFE of 1.69 for 40 primiparous cows, using the same calculation as Equation 9. Thus, application of our proposed Cal method within the RuFaS model to simulate individual farms is deemed to result in reliable farm-specific indicators of feed use and GFE. Dry matter intake and GFE values were substantially different between methods for several farms. Specifically, 6 farms (F3, F4, F7, F8, F9, F10) had DMI differences exceeding 1.5 kg/cow per day, and 6 farms (F3, F6, F7, F8, F9, F10) exhibited GFE differences greater than 0.2. Notably, for F7 and F9, DMI differences were 6.8 and 8.3 kg/cow per day, respectively, and GFE differences were 0.30 and 0.45.

Regarding manure nitrogen excretion, Cal yielded values ranging from 0.332 to 0.506 kg/cow per day, closely aligning with the reported average value of 0.422 kg/cow per day by Weiss et al. (2009). Appuhamy et al. (2018) reported average degradable volatile solids excretion rates of 5.13 kg/cow per day and 6.73 kg/cow per day from 2 separate data sets, alongside nondegradable volatile solids of 0.60 kg/cow per day and 0.88 kg/cow per day. Our estimates for degradable volatile solids ranged from 6.07 kg/cow per day to 9.30 kg/cow per day, and for nondegradable volatile solids from 0.722 kg/cow per day to 1.05 kg/cow per day, which are slightly higher than the reported values. This deviation likely stems from the higher milk production (averaging 37.5 kg/cow per day) and DMI (averaging 25.7 kg/cow per day) of our case study farms compared with the data sets in their study, which reported milk production at 23.6 kg/cow per day and 32.4 kg/cow per day, with corresponding DMI of 16.4 kg/cow per day and 21.2 kg/cow per day.

The variations in milk production, DMI, and manure excretion also led to differences in GHG emissions between the 2 methods (Table 4). Eight farms (excluding F1 and F2) had GHG intensity differences greater than 0.10 kg CO₂-eq/kg milk, among which 3 (F6, F7, F9) exhibited differences greater than 0.25 kg CO₂-eq/kg milk. Enteric CH₄ production across our case study farms ranged from 0.389 to 0.493 kg/cow per day, a range consistent with values reported in existing literature: 0.369 kg CH₄/cow per day (Niu et al., 2018), 0.361 kg CH₄/cow per day (Kebreab et al., 2023), and 0.441 kg CH₄/cow per day (Fresco et al., 2023). Estimates for CO₂-eq emissions from feeds ranged from 9.27 to 15.1 kg CO₂-eq/cow per day. These estimates align with those reported by Henriksson et al. (2014), which ranged from 11.3 to 14.4 kg CO₂-eq/cow per day. In terms of GHG emissions from manure, the predicted CH₄ and N₂O emissions based on manure excretion estimates from

our simulations ranged from 0.229 to 1.09 kg CH₄/cow per day and 1.07×10^{-3} to 1.59×10^{-3} kg N₂O/cow per day, respectively. A review study reported CH₄ and N₂O emissions from slurry storage at 0.277 ± 0.129 kg CH₄/cow per day and 0.00082 ± 0.00082 kg N₂O/cow per day, respectively, which was on a comparable scale with our findings (Owen and Silver, 2015). The total GHG intensity estimates informed by Cal ranged from 0.751 to 1.43 kg CO₂-eq/kg milk. For comparison, Nijdam et al. (2012) reviewed 12 global studies and reported values ranging from 1 to 2 kg CO₂-eq/kg milk. In the context of US dairy farms, Capper and Cady (2020) reported an intensity of 1.7 kg CO₂-eq/kg ECM, whereas Rotz et al. (2021) reported a mean of 1.04 kg CO₂-eq/kg FPCM, with a range of 0.69 to 1.45 kg CO₂-eq/kg FPCM for the cradle-to-farm gate emissions. Our results align closely with these findings, indicating consistency with reported values in the literature. On average, the percentage difference in total GHG intensity estimation between Cal and Base was 16.1% (average absolute difference of 0.18 kg CO₂-eq/kg milk, with a range of 0.000 to 0.483 kg CO₂-eq/kg milk), underscoring the significant influence of the lactation curve parameters on carbon footprints.

Overall, our results demonstrated that the Cal method within the RuFaS model achieves high accuracy in simulating farm-specific lactation curves, as evidenced by the strong agreement between simulated and observed AHMP. Direct measurement of other performance and environmental outputs are not currently available in most commercial farm settings. However, given the critical role of accurate milk production estimates in dairy farm simulation models, we expect these outcomes to be more accurate when using Cal as well. This conclusion is further supported by the close alignment of our predictions with established values in the literature. We acknowledge the utility of the Base method for understanding general trends in dairy farm performance at a regional or temporal scale. However, the Cal method offers greater versatility for individual farm applications, making it the preferred approach within the RuFaS framework. Furthermore, the Cal method has the potential beyond RuFaS, a generalized calibration method for estimating lactation curve parameters in other contexts that use the Wood's lactation curve, including alternative dairy farm simulation models.

Limitations and Future Work

Although we were able to demonstrate improved accuracy in the prediction of AHMP, the absence of measured daily milk production, DMI, manure excretion, and GHG emissions limited our ability to directly compare and conclusively determine improved accuracy of these production and environmental outcomes. Furthermore, be-

cause our method builds on the work of Li et al. (2022), which was developed and evaluated exclusively for Holsteins, application to other prevalent breeds such as Jersey should be conducted with caution. Finally, although AHMP is widely available and serves as a comprehensive indicator of whole-farm production performance, it does not fully capture the nuances of individual cow lactation dynamics, limiting the method's accuracy at the cow level. Incorporating individual animal test-day or even daily milk production data would enhance the precision of lactation curve parameters, enabling more accurate predictions at both herd and individual cow levels in farm simulation models. However, this approach would require additional data collection and a more complex computational framework, introducing new challenges for implementation.

Future research should aim to address data accessibility, integration, and computational barriers to incorporating more granular and diverse data sources for use in whole-farm simulation models and environmental footprinting analysis. An area for exploration is to connect the RuFaS model with dairy farm data integration platforms such as Dairy Brain (Cabrera et al., 2020; Ferris et al., 2020; Wangen et al., 2021) and the Cornell Agricultural Systems Testbed and Demonstration Site (<https://cals.cornell.edu/cast-farm-future>). Moreover, with the availability of more detailed data, the use of more complex lactation models—such as MilkBot or the Dijkstra model, which have demonstrated improved performance under certain conditions—could be explored (Dijkstra et al., 1997; Val-Arreola et al., 2004; Ehrlich, 2011; Cole et al., 2012). Expanding the method evaluation to include other dairy breeds and farms from different regions with different management practices will also enhance the robustness and generalizability of our proposed method.

CONCLUSIONS

This study introduces a novel method for refining parity-specific parameters of the Wood's lactation curve to improve the accuracy of AHMP estimates in dairy farm simulation models. By incorporating observed AHMP as an input, our calibration method significantly improved predicted milk production accuracy over the existing literature-based method. Through comparison of production outcomes predicted by simulations that were informed by different methods for parameterization of the lactation curves and environmental outcomes calculated based on the simulation results, we highlighted the effects of these model inputs on key downstream indicators such as GFE, manure excretion, and GHG emissions. By integrating easily accessible farm data, our method provides improved AHMP predictions, enhances the reliability of estimates for performance and environmental

indicators, and simplifies the application of dairy farm simulation models such as RuFaS.

NOTES

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Nonstandard abbreviations used: AHMP = annual herd milk production; Base = baseline lactation curve parameter approach; Cal = farm-specific calibrated lactation curve parameter approach; CO₂-eq = carbon dioxide equivalents; DNB = do-not-breed status; ED = estrus detection; F1–F10 = farm numbers for observational data collection (New York: F1–F3, Texas: F4–F6, Wisconsin: F7–F10); FoodS³ = Food System Supply-chain Sustainability model; FPCM = fat- and protein-corrected milk; GFE = gross feed efficiency; M305 = 305-d milk production; MAPE = mean absolute percentage error; RMSPE = root mean square percentage error; RuFaS = Ruminant Farm Systems model; TAI = timed artificial insemination.

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APPENDIX

Table A1. Ingredient composition of the regional lactating cow diets

Item, % of DM	Regional lactating cow diet		
	NY	WI	TX
Corn silage	39.3		23.3
Corn silage, immature		37.5	
Grass silage	7.53		5.50
Legume hay			18.0
Legume silage	13.7	17.7	
Grass/legume mix silage		0.370	
Blood meal		0.740	
Corn grain	13.7	15.3	19.7
Mineral mix	3.39	3.27	3.39
Northeast regional byproduct mix ¹	22.3		
Midwest regional byproduct mix ¹		25.2	
Western regional byproduct mix ¹			30.1

¹Regional byproduct mixes are location-specific. The top 5 ingredients for each region are as follows. Northeast (NY): soybean meal, canola meal, whey acid, treated soybean meal, and dry corn distillers grains. Midwest (WI): corn cannery waste, canola meal, cereal, soybean meal, and dry corn distillers grains. Western (TX): canola meal, wet and dry corn distillers grains, almond hulls, and cottonseed whole meal (Asselin-Balencçon et al., 2013; Thoma et al., 2013; de Ondarza and Tricarico, 2021).

Table A2. Nutritional composition of the regional lactating cow diets

Item, ¹ % of DM	Regional lactating cow diet		
	NY	WI	TX
CP	16.3	15.8	16.4
aNDF ²	33.9	34.0	32.2
ADF	21.1	21.1	20.9
Lignin	3.48	3.66	4.44
Starch	24.1	24.4	25.2
Ether extract	3.82	4.05	4.59

¹The nutritional composition aligns with diets formulated using data from over 2,000 farms collected through the FARM-ES survey, supplemented by recent study findings (Asselin-Balencçon et al., 2013; Thoma et al., 2013; de Ondarza and Tricarico, 2021).

²aNDF = amylase-treated NDF.

Table A3. Herd parity composition estimates ($\pm$ SD) using baseline (Base) and calibrated (Cal) methods for 10 dairy farms in New York (F1–F3), Texas (F4–F6), and Wisconsin (F7–F10)

Farm	Method	Parity 1 (%)	Parity 2 (%)	Parity 3+ (%)
F1	Base	36.0 $\pm$ 0.6	26.3 $\pm$ 0.9	37.7 $\pm$ 0.4
	Cal	35.0 $\pm$ 0.3	25.4 $\pm$ 0.8	39.7 $\pm$ 0.8
F2	Base	34.6 $\pm$ 1.1	23.5 $\pm$ 0.6	41.9 $\pm$ 0.8
	Cal	34.0 $\pm$ 0.7	22.6 $\pm$ 1.0	43.3 $\pm$ 1.1
F3	Base	26.9 $\pm$ 4.3	23.9 $\pm$ 5.5	49.1 $\pm$ 8.4
	Cal	23.9 $\pm$ 5.0	20.8 $\pm$ 6.9	55.2 $\pm$ 8.5
F4	Base	40.3 $\pm$ 0.9	27.3 $\pm$ 0.6	32.4 $\pm$ 0.7
	Cal	38.2 $\pm$ 0.4	27.3 $\pm$ 0.4	34.5 $\pm$ 0.6
F5	Base	47.7 $\pm$ 0.8	27.7 $\pm$ 0.6	24.5 $\pm$ 0.9
	Cal	45.8 $\pm$ 0.4	26.5 $\pm$ 0.3	27.6 $\pm$ 0.2
F6	Base	43.0 $\pm$ 0.5	27.9 $\pm$ 0.1	29.1 $\pm$ 0.5
	Cal	42.6 $\pm$ 0.2	27.6 $\pm$ 0.2	29.9 $\pm$ 0.2
F7	Base	40.0 $\pm$ 4.3	24.7 $\pm$ 2.5	35.3 $\pm$ 3.7
	Cal	38.0 $\pm$ 6.8	23.7 $\pm$ 4.2	38.3 $\pm$ 6.2
F8	Base	38.0 $\pm$ 1.3	25.5 $\pm$ 1.7	36.5 $\pm$ 1.0
	Cal	37.8 $\pm$ 2.1	24.9 $\pm$ 1.0	37.3 $\pm$ 2.2
F9	Base	32.1 $\pm$ 2.3	30.2 $\pm$ 6.2	37.6 $\pm$ 7.3
	Cal	39.6 $\pm$ 5.6	27.6 $\pm$ 5.2	32.7 $\pm$ 0.7
F10	Base	31.1 $\pm$ 0.6	23.4 $\pm$ 0.4	45.5 $\pm$ 0.6
	Cal	30.5 $\pm$ 1.6	23.1 $\pm$ 0.5	46.4 $\pm$ 1.0